



```

name: <unnamed>
log: D:\Financial Modeling\Bahria Fall 2025\Panel Data\Stata Results.smcl
log type: smcl
opened on: 20 Dec 2025, 15:06:10

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1 . encode company, gen(comp)

2 .

3 . tsset comp year
    panel variable: comp (unbalanced)
    time variable: year, 2007 to 2010, but with gaps
                  delta: 1 unit

4 .

5 . reg cash cft liq lev mbr size

```

Source	SS	df	MS	Number of obs =	1005
Model	3.89217968	5	.778435935	F(5, 999) =	128.88
Residual	6.03400811	999	.006040048	Prob > F =	0.0000
				R-squared =	0.3921
				Adj R-squared =	0.3891
Total	9.92618778	1004	.009886641	Root MSE =	.07772

cash	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
cft	.0498717	.0128417	3.88	0.000	.0246718	.0750715
liq	.0045446	.001186	3.83	0.000	.0022174	.0068719
lev	.0045336	.0011858	3.82	0.000	.0022066	.0068606
mbr	.0018393	.0004299	4.28	0.000	.0009956	.0026829
size	.0069457	.0014679	4.73	0.000	.0040652	.0098263
_cons	-.071728	.0218903	-3.28	0.001	-.1146843	-.0287716

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6 .

7 . hettest

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Breusch-Pagan / Cook-Weisberg test for heteroskedasticity
Ho: Constant variance
Variables: fitted values of cash

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chi2(1)      =      3.41
Prob > chi2   =      0.0648

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8 .

9 . ssc install winsor
   checking winsor consistency and verifying not already installed...
   all files already exist and are up to date.

10 .

11 . winsor cash,gen (wcash) p(0.1)

12 .

13 . winsor cft, gen (wcf) p(0.1)

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14 .
15 . winsor liq , gen (wliq) p(0.1)

16 .
17 . winsor lev , gen (wlev) p(0.1)

18 .
19 . winsor mbr , gen (wmbr) p(0.1)

20 .
21 . winsor size , gen (wsz) p(0.1)

22 .
23 . reg wcash wcf wliq wlev wmbr wsz

```

Source	SS	df	MS	Number of obs =	1005
Model	.434752079	5	.086950416	F(5, 999) =	99.52
Residual	.872783669	999	.000873657	Prob > F =	0.0000
Total	1.30753575	1004	.001302326	R-squared =	0.3325
				Adj R-squared =	0.3292
				Root MSE =	.02956

wcash	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
wcf	.0584347	.0131636	4.44	0.000	.0326032	.0842662
wliq	.0413024	.0048434	8.53	0.000	.0317979	.0508069
wlev	-.0435425	.0050055	-8.70	0.000	-.0533651	-.03372
wmbr	.024437	.0031652	7.72	0.000	.0182259	.0306482
wsz	.0025941	.0006801	3.81	0.000	.0012595	.0039287
_cons	-.0290959	.0104139	-2.79	0.005	-.0495314	-.0086603

```

24 .
25 . xtreg wcash wcf wliq wlev wmbr wsz, fe

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Fixed-effects (within) regression      Number of obs   =      1005
Group variable: comp                   Number of groups =       313

R-sq:  within = 0.0515                  Obs per group:  min =        1
      between = 0.2769                      avg   =       3.2
      overall  = 0.2296                      max   =        4

corr(u_i, Xb) = 0.3268                  F(5, 687)       =       7.47
                                          Prob > F         =      0.0000

```

wcash	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
wcf	.0166351	.0136069	1.22	0.222	-.010081	.0433512
wliq	.0050993	.0066619	0.77	0.444	-.0079808	.0181794
wlev	-.0242278	.0054481	-4.45	0.000	-.0349248	-.0135308
wmbr	.0108332	.0038144	2.84	0.005	.0033438	.0183225
wsz	-.0017981	.0038221	-0.47	0.638	-.0093024	.0057062
_cons	.0490643	.0573659	0.86	0.393	-.0635691	.1616977
sigma_u	.02866038					
sigma_e	.01969956					
rho	.67914358	(fraction of variance due to u_i)				

```

F test that all u_i=0:      F(312, 687) =      5.01      Prob > F = 0.0000

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26 .
27 . estimates store fe

28 .
29 . xtreg wcash wcf wliq wlev wmbrr wsz, re

```

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Random-effects GLS regression           Number of obs   =      1005
Group variable: comp                   Number of groups =       313

R-sq:  within = 0.0455                 Obs per group: min =        1
        between = 0.4236                  avg =       3.2
        overall = 0.3288                  max =        4

corr(u_i, X) = 0 (assumed)              Wald chi2(5)      =     215.52
                                           Prob > chi2       =      0.0000

```

wcash	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
wcf	.0390773	.0120742	3.24	0.001	.0154122	.0627423
wliq	.0287903	.0050872	5.66	0.000	.0188195	.038761
wlev	-.036974	.0046474	-7.96	0.000	-.0460827	-.0278653
wmbrr	.018707	.0030723	6.09	0.000	.0126855	.0247286
wsz	.0030968	.0009838	3.15	0.002	.0011686	.005025
_cons	-.0303345	.0148264	-2.05	0.041	-.0593938	-.0012753
sigma_u	.02106678					
sigma_e	.01969956					
rho	.53350035	(fraction of variance due to u_i)				

```

30 .
31 . estimates store re

32 .
33 . hausman fe re

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	Coefficients		(b-B) Difference	sqrt(diag(V_b-V_B)) S.E.
	(b) fe	(B) re		
wcf	.0166351	.0390773	-.0224422	.0062739
wliq	.0050993	.0287903	-.023691	.0043013
wlev	-.0242278	-.036974	.0127462	.0028433
wmbrr	.0108332	.018707	-.0078739	.0022607
wsz	-.0017981	.0030968	-.0048949	.0036933

b = consistent under Ho and Ha; obtained from xtreg
B = inconsistent under Ha, efficient under Ho; obtained from xtreg

Test: Ho: difference in coefficients not systematic

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chi2(5) = (b-B)' [(V_b-V_B)^(-1)] (b-B)
        =      51.22
Prob>chi2 =      0.0000

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34 .

35 . xtabond wcash wcf wliq wlev wmbrr wsz, lags(1) twostep artests(2)

Arellano-Bond dynamic panel-data estimation Number of obs = **282**
 Group variable: **comp** Number of groups = **143**
 Time variable: **year**

Obs per group: min = **1**
 avg = **1.972028**
 max = **2**

Number of instruments = **9** Wald chi2(6) = **8.56**
 Prob > chi2 = **0.2002**

Two-step results

wcash	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
wcash L1.	.310204	.2519381	1.23	0.218	-.1835855	.8039936
wcf	-.0008969	.0234733	-0.04	0.970	-.0469036	.0451099
wliq	.0031041	.0126718	0.24	0.806	-.0217322	.0279404
wlev	.0034655	.0103782	0.33	0.738	-.0168754	.0238063
wmbrr	.0135379	.0085445	1.58	0.113	-.0032091	.0302848
wsz	-.0181227	.0099059	-1.83	0.067	-.037538	.0012926
_cons	.2752608	.1507462	1.83	0.068	-.0201964	.570718

Warning: gmm two-step standard errors are biased; robust standard errors are recommended.

Instruments for differenced equation

GMM-type: **L(2/.) .wcash**Standard: **D.wcf D.wliq D.wlev D.wmbrr D.wsz**

Instruments for level equation

Standard: **_cons**

36 .

37 . xtabond wcash wcf wliq wlev wmbrr wsz, inst(wcf wlev) lags(1) maxlags(1) twostep artests(2)

Arellano-Bond dynamic panel-data estimation Number of obs = **282**
 Group variable: **comp** Number of groups = **143**
 Time variable: **year**

Obs per group: min = **1**
 avg = **1.972028**
 max = **2**

Number of instruments = **11** Wald chi2(6) = **16.57**
 Prob > chi2 = **0.0110**

Two-step results

wcash	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
wcash L1.	.5307774	.145297	3.65	0.000	.2460006	.8155543
wcf	.0082402	.024703	0.33	0.739	-.0401769	.0566573
wliq	-.0036292	.0128651	-0.28	0.778	-.0288443	.0215859
wlev	.0157472	.0073785	2.13	0.033	.0012856	.0302089
wmbrr	.014097	.0094298	1.49	0.135	-.0043851	.0325791
wsz	-.0122145	.0107305	-1.14	0.255	-.0332459	.0088169
_cons	.1760321	.1631485	1.08	0.281	-.1437331	.4957973

Warning: gmm two-step standard errors are biased; robust standard errors are recommended.

Instruments for differenced equation

GMM-type: **L(2/.) .wcash**Standard: **D.wcf D.wliq D.wlev D.wmbrr D.wsz wcf wlev**

Instruments for level equation

Standard: **_cons**

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38 .
39 . estat sargan
    Sargan test of overidentifying restrictions
      H0: overidentifying restrictions are valid

      chi2(4)      =    4.265647
      Prob > chi2   =    0.3712
```

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40 .
41 . estat abond
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Arellano-Bond test for zero autocorrelation in first-differenced errors

Order	z	Prob > z
1	-2.8113	0.0049
2	.	.

H0: no autocorrelation

```
42 . log close
    name: <unnamed>
    log:  D:\Financial Modeling\Bahria Fall 2025\Panel Data\Stata Results.smcl
    log type: smcl
    closed on: 20 Dec 2025, 15:07:30
```
